

CR-Touch → Educational Macro-SPM Z-Probe Project (Final Version)

1. Project Overview

This project converts a Creality CR-Touch auto-leveling sensor into a contact-based Z probe for macro-scale surface scanning. The system demonstrates core principles of scanning probe microscopy (SPM) using low-cost, accessible components. Two probe tip configurations are supported: a rigid steel needle (recommended for stability) and a pogo-pin probe (optional, safer but less precise).

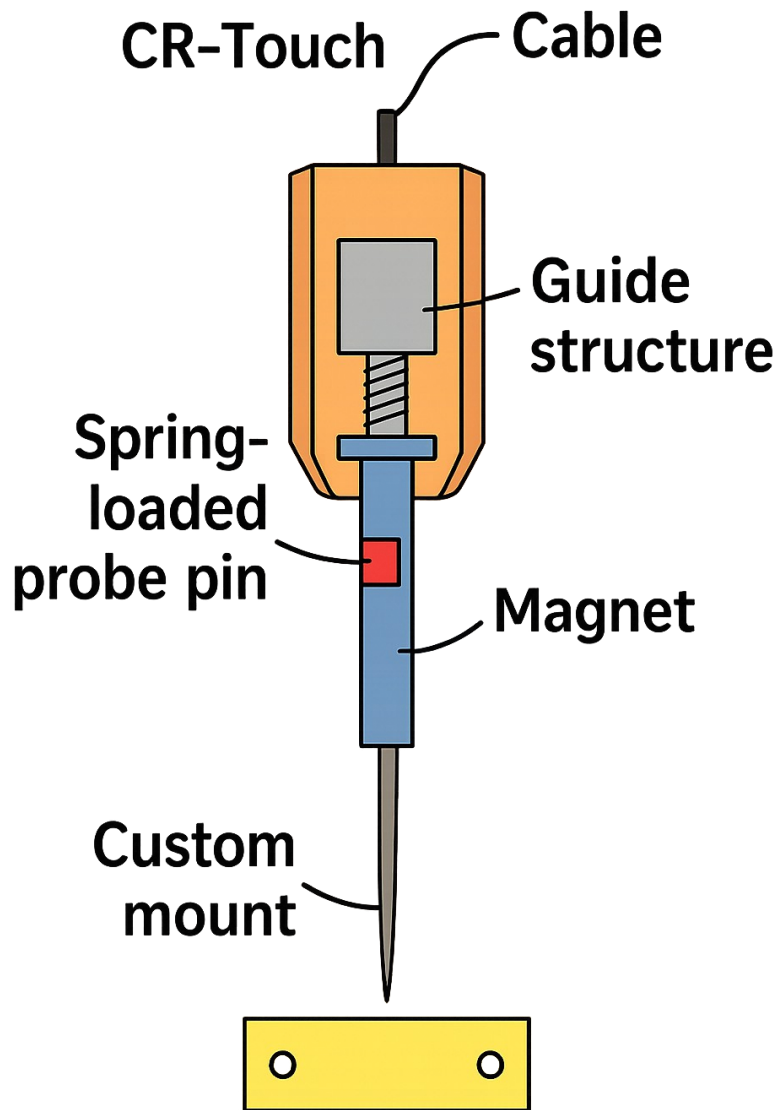


Figure: System overview (sensor, guide, probe tip, magnet)

2. 1. Working Principle

The CR-Touch contains a spring-loaded probe and a small internal magnet. A Hall-effect sensor detects magnet displacement and a microcontroller outputs a digital trigger. Replacing the probe tip with a needle or pogo pin converts the device into a precise contact probe. The added external guide improves vertical stability and repeatability.

3. 2. Bill of Materials (BOM)

REQUIRED PARTS:

- Creality CR-Touch sensor
- Neodymium magnet 2x2 mm (mandatory, pogo pins are NOT magnetic)
- PTFE tube 2 mm OD / 1 mm ID
- M3 screws

PROBE OPTIONS:

Option A (Recommended): Steel needle (~1 mm diameter)

Option B (Optional): Pogo pin (P50-B1 or P75-B1, spring-loaded, requires glued magnet)

3D PRINTED PARTS:

- CR-Touch mount
- Guide tube holder
- Calibration shims

4. 3. Probe Tip Options (Important Decision)

Steel Needle:

- High repeatability
- Single spring system (CR-Touch only)
- Sharp tip for better spatial resolution

Pogo Pin (P50/P75):

- Built-in spring (safer contact)
- Requires external magnet attachment
- Creates double-spring system -> lower measurement precision

5. 4. CAD and Mechanical Design

- Guide hole: 2.0–2.1 mm
- Guide length: 10–15 mm
- Mount must be rigid
- Probe axis must be perfectly vertical
- Ensure smooth motion with minimal friction

6. 5. Assembly Instructions

1. Remove original CR-Touch probe pin
2. Prepare probe:
 - Needle: cut to ~18 mm
 - Pogo pin: use full length
3. Glue magnet to top (centered and aligned)

4. Insert with original spring
5. Print and install guide + PTFE tube
6. Mount to printer using printed mount
7. Align probe vertically

7. 6. Testing Procedure

- Power CR-Touch
- Press probe manually
- Verify consistent trigger point
- If unstable: adjust magnet position
- Ensure smooth motion without sticking

8. 7. Safety Notes

- Needle tips are sharp → handle carefully
- Use slow Z movement
- Avoid hard collisions
- Keep magnets away from electronics

9. 8. Summary

This system provides a low-cost, educational Z-probe solution. For best performance use a steel needle. For safer contact a pogo pin can be used, but requires compromises in precision.